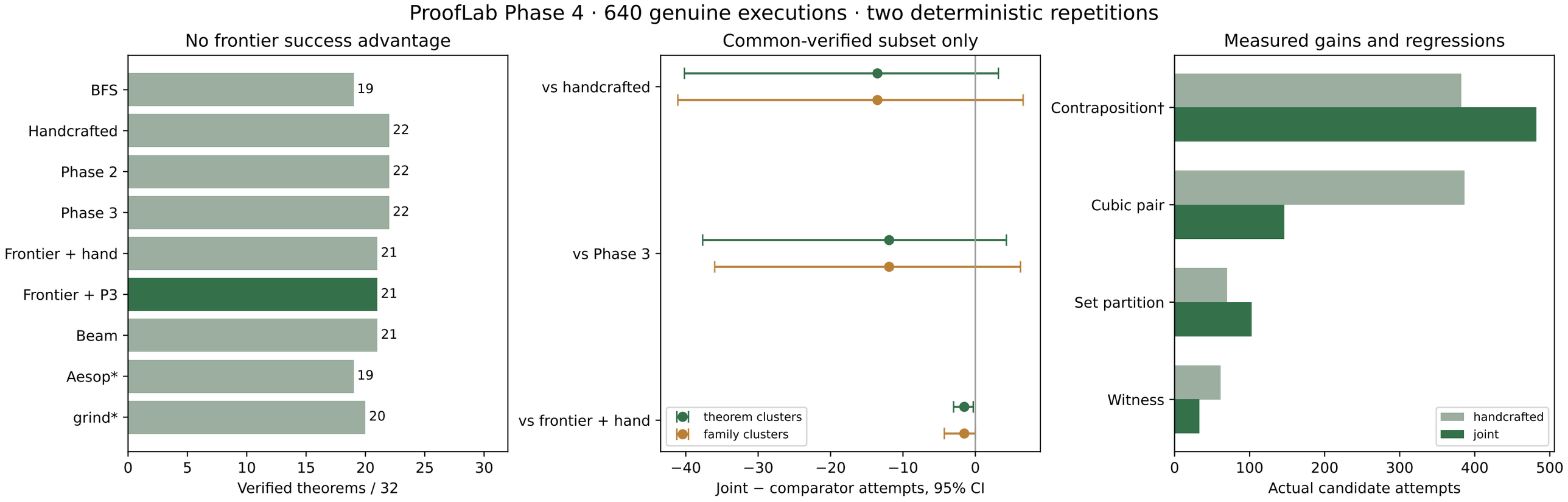
ProofLab Phase 4 · 640 genuine executions · two deterministic repetitions
No frontier success advantage
BFS
Handcrafted
Phase 2
Phase 3
Frontier + hand
Frontier + P3
Beam
Aesop*
grind*
19
22
22
22
21
21
21
19
20
0
5
10
15
20
25
30
Verified theorems / 32
Common-verified subset only
vs handcrafted
vs Phase 3
vs frontier + hand
theorem clusters
family clusters
−40
−30
−20
−10
0
Joint − comparator attempts, 95% CI
Measured gains and regressions
Contraposition†
Cubic pair
Set partition
Witness
handcrafted
joint
0
100
200
300
400
500
Actual candidate attempts
* Native references have incomparable internal work. † Joint fails; fewer/more failed-run attempts are not efficiency gains.